Console : install.packages("lubridate")

as.character( )

as.numeric( )

library(lubridate)

Dates

~~as.date~~

Introduction to lubridate

lubridate is a popular R package that makes it easy to work with date-times. It provides a wide range of functions and methods to parse, manipulate, and format date-time objects. This tutorial will cover some of the fundamental aspects of using lubridate for working with date and time data in R.

Creating Date-Time Objects

1. ymd(): is a function used to create a date object from year, month, and day components.

```
my_date <- "2023-09-15"
class(my_date)
my_date <- ymd(my_date)
class(my_date)
```

Output: Character

Output: Date

2. mdy(): is a function used to create a date object in a month-day-year format.

```
my_date2 <- "12-14-2023"
my_date2 <- mdy(my_date2)
my_date2
```

Output: "2023-12-14"

3. ymd_hms(): creates a date-time object with hours, minutes, and seconds.

```
my_datetime <- "2023-09-15 14:30:45"
my_datetime <- ymd_hms(my_datetime)
```

→ also defines class date

Extracting Components

4. Extracting Year, Month, Day, etc.

```
my_date <- ymd("2024-10-30")
```

converts to class date

Extract year

```
year(my_date)
```

Output: 2024

Extract month

```
month(my_date)
```

Output: 10

Extract day

```
day(my_date)
```

Output: 30

are numeric

Extract day of the week (Sunday: 1, Monday: 2, etc.)

```
wday(my_date)
```

Output: 4 (Wednesday)

numeric

1 – Sunday
2 – Monday
⋮
7 – Sat

Example with a Dataframe

dwts_dates

Name	Birth	Wedding
Derek	1985-05-17	08-26-2023
Mark	1986-05-24	11-25-2016
Lindsay	1994-01-11	06-18-2015

y m d m d y

```
class(dwts_dates$Name)
```

Output: Character

```
class(dwts_dates$Birth)
```

Output: Character

```
class(dwts_dates$Wedding)
```

Output: Character

overwrite

```
dwts_dates <- dwts_dates %>%
  mutate(Birth = ymd(Birth),
         Wedding = mdy(Wedding))
```

Same col names, overwrite columns

Name	Birth	Wedding
Derek	1985-05-17	2023-08-26
Mark	1986-05-24	2016-11-25
Lindsay	1994-01-11	2015-06-18

```
class(dwts_dates$Name)
```
Output: Character

```
class(dwts_dates$Birth)
```
Output: Date

```
class(dwts_dates$Wedding)
```
Output: Date

```
dwts_info <- dwts_dates %>%
  mutate(Year = year(Birth),
         Day = day(Wedding),
         Month = month(Wedding),
         Day_Week = wday(Wedding))
```

Name	Birth	Wedding	Year	Day	Month	Day-week
Derek	1985-05-17	2023-08-26	1985	26	08	7
Mark	1986-05-24	2016-11-25	1986	25	11	6
Lindsay	1994-01-11	2015-06-18	1994	18	06	5

Arithmetic with Date-Times

5. Adding Days

```
my_date <- ymd("2024-10-30")
new_date <- my_date + days(7)  # Add 7 days to my_date
new_date
```

Output: 2024-11-06

years(7)
months()

6. Finding Time Difference

```
my_date <- ymd("2024-10-30")
my_date2 <- ymd("2024-11-06")
```

Find the time difference between my_date and my_datetime

```
diff <- my_date2 - my_date
diff
```

Output: Time difference of 7 days

Use difftime when the date includes time

```
time1 <- ymd_hms("2023-09-15 08:30:00")
time2 <- ymd_hms("2023-09-15 12:45:30")
```

months / mins

- ***To calculate the time difference in seconds***

```
difftime(time2, time1, units = "secs")
```

Output: Time difference of 15330 secs.

```
as.numeric(difftime(time2, time1, units = "secs"))
```

Output: 15330

- ***Calculate the time difference in hours***

```
difftime(time2, time1, units = "hours")
```

Output: Time difference of 4.258333 hours

```
as.numeric(difftime(time2, time1, units = "hours"))
```

Output: 4.258333

Formatting Dates

7. Using `format()`: you can format date-time objects for display

```
my_date <- ymd("2026-10-26")
```

vector, command

```
format(my_date, format = "%A")  # %A - The day of the week (i.e, Friday)
```

Output: Monday

```
format(my_date, format = "%a") # %a - The day of the week truncated (i.e, Fri)
```

Output: Mon

```
format(my_date, format = "%B")  # %B - The full name of the month (i.e September)
```

Output: October

```
format(my_date, format = "%b")  # %b - The name of the month truncated (i.e., Sep)
```

Output: Oct

```
format(my_date, format = "%D")  # %D - The date in month/day/year format.
```

Output: 10/26/2026

```
format(my_date, format = "%d") # %d - The number of the day in month
```

Output: 26

```
format(my_date, format = "%Y") # %Y - The year with four digits
```

Output: 2026

```
format(my_date, format = "%y") # %y - The year with two digits
```

Output: 26

```
format(my_date, format = "%m")  # %m - The number of the month
```

Output: 10

Suppose you want it to be in the format "Friday, September 15, 2023"

```
format(my_date, format = "%A, %B %d, %Y")
```

Output: "Monday, October 26, 2026"